

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

1. Administrative & Study Identification

Full Study Title:	A Phase 4 Study to Evaluate Safety and Effectiveness of Luspatercept (ACE-536) for the Treatment of Anemia Due to IPSS-R Very Low, Low, or Intermediate Risk Myelodysplastic Syndromes (MDS) With Ring Sideroblasts Who Require Red Blood Cell Transfusions in Participants Who Have Had Unsatisfactory Response to or Are Ineligible to Erythropoietin Based Therapy and in Participants With Transfusion Dependent Anemia Due to Beta-Thalassemia
Short Title:	A Study to Evaluate the Safety and Effectiveness of Luspatercept for the Treatment of Transfusion-dependent (TD) Anemia Associated With Myelodysplastic Syndromes (MDS) & Beta-thalassemia (?-Thal) in India
Protocol Number:	716558171765213578
NCT Number:	NCT05891249
Sponsor Name:	Bristol Myers Squibb
Investigational Product:	luspatercept
Phase:	PHASE4
Medical Monitor:	[Not Provided in Posting]

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective:	1. ?-Thal Cohort: Number of participants with treatment-related adverse events (AEs) of grade 3 or higher 2. MDS-Ring Sideroblasts (RS) Cohort: Number of participants with treatment-related AEs of grade 3 or higher
Secondary Obj.:	1. ?-Thal Cohort: Percentage of participants who achieved red blood cell (RBC) transfusion burden reduction (? 33% reduction from baseline) with a reduction of at least 2 red cell units compared to the 12-week interval prior to enrollment 2. ?-Thal Cohort: Percentage of participants who achieved RBC transfusion burden reduction of at least 33% from baseline during any 12-week interval with a reduction of at least 2 red cell units compared to the 12-week interval prior to enrollment 3. MDS-RS Cohort: Percentage of participants who achieved RBC-TI during any consecutive 56-day period

2.2 Background & Rationale

The purpose of this study is to evaluate the safety and effectiveness of luspatercept in participants who require regular blood cell transfusions due to b-thalassemia and myelodysplastic syndromes in India

3. Study Design & Methodology

Design Framework:	Type: INTERVENTIONAL, Phase: PHASE4, Status: ACTIVE_NOT_RECRUITING
Target N:	85

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria

β-Thalassemia Cohort

- * Documented diagnosis of β-thalassemia or hemoglobin (Hb E/β-thalassemia). (β-thalassemia with mutation and/or multiplication of alpha [α] globin is allowed).
- * Regularly transfused, defined as 6 RBC units to 20 RBC units in the 24 weeks prior to enrollment and no transfusion-free period for > 35 days during that period.

MDS-RS Cohort

Participant has documented diagnosis of MDS according to World Health Organization (WHO) (2016)/French-American-British FAB classification that meets revised International Prognostic Scoring System (IPSS-R) classification of very low, low, or intermediate risk disease and the following criteria: i) RS ≥ 15% of erythroid precursors in bone marrow. If the SF3B1 mutation is present, RS ≥ 5% will be included.

ii) Less than 5% blasts in bone marrow and < 1% peripheral blood blasts. iii) Peripheral blood white blood cell (WBC) count < 13,000/ microliters (μL).

- * If the participant was previously treated with erythropoiesis-stimulating agents (ESAs) or granulocyte colony-stimulating factor (G-CSF)/granulocyte-macrophage colony-stimulating factor (GM-CSF), both agents must have been discontinued ≥ 4 weeks prior to the date of enrollment.

4.2 Exclusion Criteria

β-Thalassemia Cohort

- * A diagnosis of Hb S/β-thalassemia or β-thalassemia (for example, Hemoglobin H).
- * Deep vein thrombosis (DVT) or stroke requiring medical intervention ≥ 24 weeks prior to enrollment.
- * Use of chronic anticoagulant therapy is excluded unless the treatment stopped at least 28 days prior to enrollment. Anticoagulant therapies used for prophylaxis for surgery or high-risk procedures as well as low-molecular-weight (LMW) heparin for superficial venous thrombosis and chronic aspirin are allowed.
- * Cytotoxic agents or immunosuppressants or immunomodulatory drugs (IMiDs) ≥ 28 days prior to enrollment (ie, antithymocyte globulin or cyclosporine or thalidomide).

MDS-RS Cohort

- * MDS associated with del 5q cytogenetic abnormality.
- * Secondary MDS, that is, MDS that is known to have arisen as the result of chemical injury or treatment with chemotherapy and/or radiation for other diseases.
- * Participant has known clinically significant anemia due to iron, vitamin B12, or folate deficiencies; autoimmune or

Clinical Study Summary Report (CSSR)

Document Status: Final | Version: 1.0 | Date: 01-APR-2026

hereditary hemolytic anemia; or gastrointestinal bleeding.

* Iron deficiency to be determined by serum ferritin ≥ 15 micrograms per liter ($\mu\text{g/L}$) and additional testing if clinically indicated (for example, calculated transferrin saturation $\leq 20\%$ or bone marrow aspirate [BMA] stain for iron).

11. Study Identifiers & Governance

Primary ID: 716558171765213578

Registry Number: NCT05891249

Sponsor: Bristol Myers Squibb

Study Title: A Study to Evaluate the Safety and Effectiveness of Luspatercept for the Treatment of Transfusion-dependent (TD) Anemia Associated With Myelodysplastic Syndromes (MDS) & Beta-thalassemia (β -Thal) in India

15. Sign-off & Approval

Principal Investigator: _____ Date: _____

Clinical Operations Lead: _____ Date: _____

Lead Statistician: _____ Date: _____